

Siddartha Reddy Munugala

New Jersey • (636) 262-1008 • Siddarth.lbf341@gmail.com • github.com/Redsid5 • linkedin.com/in/siddartharm • vibein.us • arenahq.us

EDUCATION

University of South Dakota New Jersey • Expected May 2027

B.S. Computer Science — Minor in Innovation & Entrepreneurship
Relevant Coursework: Data Structures, Algorithms, Systems Programming, Database Systems, Software Engineering

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, SQL
Frameworks	Next.js 14, React, Node.js, Tailwind CSS
AI / ML	XGBoost, SHAP, PyTorch, Claude API, Gemini API, OpenAI API, LangChain, Whisper
Infrastructure	Supabase, PostgreSQL, Firebase, Neon Postgres, Redis, Docker, Vercel, Git, GitHub Actions
Other	WebRTC, WebSockets, REST APIs, FFmpeg, CodeMirror 6, tree-sitter, NetworkX

EXPERIENCE

Co-Founder & Lead Engineer — VibeIn / ArenaHQ Jan 2025 – Present

vibein.us • arenahq.us
Built and shipped two full-stack web applications using Next.js 14, Supabase, WebRTC, and Twilio serving a combined 1,621 registered users with 20% active retention pre-beta.
Resolved 8+ critical production bugs — including RLS authentication failures, Gemini API deprecation, broken SMS/WhatsApp notifications, and Supabase Realtime failures — under a hard deployment deadline.
Implemented row-level security, real-time chat, peer recommendation matrices, and P2P camera authentication for live exam proctoring.
Conducted 50+ user interviews to drive feature iteration; managed end-to-end Vercel deployment pipelines with 99%+ uptime across both platforms.

Production Operations Staff — USD Athletics Aug 2025 – Present

University of South Dakota • Part-Time On-Campus
Operate live-event audio systems, stadium display boards, and broadcast cues for NCAA athletics events under zero-tolerance real-time production constraints.

IT Coordinator & Publicity Chair — NSLS Executive Board Sep 2024 – Present

University of South Dakota
Built member tracking pipelines, communication scripts, and recruitment dashboards; streamlined onboarding workflows for a national collegiate leadership organization.

PROJECTS

EKO — AI Background Music & Voice Dubbing System 2026 – Present

Built a three-mode AI audio platform: Studio (vocal sketch to 10 BGM variations), BGM Dub (scene-energy-matched music generation for video), and Voice Dub (multilingual speech replacement with background music preservation via Demucs stem separation).
Stack: Next.js 14 (Vercel), FastAPI (Render), Neon Postgres / Drizzle ORM, Vercel Blob, faster-whisper, librosa, Gemini Flash, Google Lyria 3 for AI audio generation, Demucs htdemucs for stem separation.
Resolved critical memory constraints on 2GB Render container by migrating from local MusicGen-small (1.8GB RAM peak) to HuggingFace Inference API, dropping footprint to ~400MB; diagnosed and fixed silent loop failures causing single-variation generation bug.
821 registered users, 40% DAU at zero paid acquisition cost; arXiv preprint (cs.SD) drafted covering full system architecture, unit economics, and competitive differentiation.

PRISM — PR Risk Intelligence & Structural Modeling 2026

Engineered a machine learning pipeline predicting post-merge engineering risk using 45 features (AST complexity via tree-sitter, dependency graphs via NetworkX, code churn); XGBoost classifier validated at AUC ≥ 0.80 with SHAP-based explainability.
Maintained cross-repository generalization gap ΔAUC < 0.05; authored formal research proposals with mathematical appendices for USD faculty and external mentorship review.

COGN — Cognitive Drift Detection GitHub Action 2026

Built an open-source CI/CD GitHub Action that scores pull requests for "cognitive drift" — measurable shifts in reasoning coherence introduced by AI-generated code — against repository baselines using three custom primitives.

ALASKA — Encrypted AI Assistant for Military Veterans 2026

Built in 48 hours: AES-256-GCM encrypted AI assistant with VA routing algorithm, crisis detection, and accessibility modes — top-10 hackathon finish.

LEADERSHIP & ACTIVITIES

AI Education Workshop Facilitator Ongoing

Self-Initiated • Youth Outreach
Design and lead hands-on workshops teaching teenagers (ages 13–16) how AI integrates into sectors of daily life — including healthcare, education, media, and transportation.
Build curriculum that makes applied AI concepts accessible to non-technical learners with no prior background.
Community Educator (Self-Initiated, Offline) — go into public spaces to teach teenagers (ages 13–16) foundational life and entrepreneurship skills: what a business is, how to find customers, how to build a product, and how to communicate and present ideas with confidence.
USD INFORMS (Beacom School of Business) — authored 10-week strategic operational plan; built chapter admin portal.
NSLS Executive Board — IT Coordinator & Publicity Chair; USD Marketing Club member.